



qualcomm/**LiteHRNet**

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Paper for qualcomm/LiteHRNet

Lite-HRNet: A Lightweight High-Resolution Network

Paper • 2104.06403 • Published Apr 13, 2021



🔗 LiteHRNet: Optimized for Qualcomm Devices

LiteHRNet is a machine learning model that detects human pose and returns a location and confidence for each of 17 joints.

This is based on the implementation of LiteHRNet found [here](#). This repository contains pre-exported model files optimized for Qualcomm® devices. You can use the [Qualcomm® AI Hub Models](#) library to export with custom configurations. More details on model performance across various devices, can be found [here](#).

Qualcomm AI Hub Models uses [Qualcomm AI Hub Workbench](#) to compile, profile, and evaluate this model. [Sign up](#) to run these models on a hosted Qualcomm® device.

🔗 Getting Started

There are two ways to deploy this model on your device:

🔗 Option 1: Download Pre-Exported Models

Below are pre-exported model assets ready for deployment.

Runtime	Precision	Chipset	SDK Versions	Download
ONNX	float	Universal	QAI RT 2.45, ONNX Runtime 1.27.1	Download
QNN_DLC	float	Universal	QAI RT 2.45	Download
TFLITE	float	Universal	QAI RT 2.45	Download

For more device-specific assets and performance metrics, visit [LiteHRNet on Qualcomm® AI Hub](#).

Option 2: Export with Custom Configurations

Use the [Qualcomm® AI Hub Models](#) Python library to compile and export the model with your own:

- Custom weights (e.g., fine-tuned checkpoints)
- Custom input shapes
- Target device and runtime configurations

This option is ideal if you need to customize the model beyond the default configuration provided here.

See our repository for [LiteHRNet on GitHub](#) for usage instructions.

Model Details

Model Type: Model_use_case.pose_estimation

Model Stats:

- Input resolution: 256x192

- Number of parameters: 1.11M
- Model size (float): 4.49 MB

 Performance Summary

Model	Runtime	Precision	Chipset	Inference Time (ms)	Peak Memory Range (MB)	Primary Compute Unit
LiteHRNet	ONNX	float	Snapdragon® X2 Elite	2.842 ms	2 - 2 MB	NPU
LiteHRNet	ONNX	float	Snapdragon® X Elite	5.61 ms	5 - 5 MB	NPU
LiteHRNet	ONNX	float	Snapdragon® 8 Gen 3 Mobile	3.063 ms	0 - 121 MB	NPU
LiteHRNet	ONNX	float	Snapdragon® 8 Gen 1 Mobile	6.228 ms	1 - 120 MB	NPU
LiteHRNet	ONNX	float	Qualcomm® Dragonwing™ QCS8550 (Proxy)	5.338 ms	0 - 123 MB	NPU
LiteHRNet	ONNX	float	Qualcomm® QCS8450	6.228 ms	1 - 120 MB	NPU
LiteHRNet	ONNX	float	Qualcomm® Dragonwing™ IQ-9075	5.679 ms	1 - 4 MB	NPU
LiteHRNet	ONNX	float	Qualcomm® Dragonwing™ IQ-X7181	5.61 ms	5 - 5 MB	NPU
LiteHRNet	ONNX	float	Qualcomm® Dragonwing™ Q-	2.835 ms	0 - 95 MB	NPU

Model	Runtime	Precision	Chipset	Inference Time (ms)	Peak Memory Range (MB)	Primary Compute Unit
			8750			
LiteHRNet	ONNX	float	Snapdragon® 8 Elite Mobile	2.835 ms	0 - 95 MB	NPU
LiteHRNet	ONNX	float	Snapdragon® 8 Elite Gen 5 Mobile	2.738 ms	0 - 96 MB	NPU
LiteHRNet	QNN_DLC	float	Snapdragon® X2 Elite	1.23 ms	1 - 1 MB	NPU
LiteHRNet	QNN_DLC	float	Snapdragon® X Elite	2.365 ms	1 - 1 MB	NPU
LiteHRNet	QNN_DLC	float	Snapdragon® 8 Gen 3 Mobile	1.346 ms	0 - 105 MB	NPU
LiteHRNet	QNN_DLC	float	Snapdragon® 8 Gen 1 Mobile	2.859 ms	0 - 103 MB	NPU
LiteHRNet	QNN_DLC	float	Qualcomm® Dragonwing™ QCS8275	4.96 ms	1 - 78 MB	NPU
LiteHRNet	QNN_DLC	float	Qualcomm® Dragonwing™ QCS8550 (Proxy)	2.062 ms	1 - 2 MB	NPU
LiteHRNet	QNN_DLC	float	Qualcomm® SA8775P	2.651 ms	1 - 80 MB	NPU
LiteHRNet	QNN_DLC	float	Qualcomm® SA8650P	2.651 ms	1 - 80 MB	NPU
LiteHRNet	QNN_DLC	float	Qualcomm® SA8255P	2.651 ms	1 - 80 MB	NPU

Model	Runtime	Precision	Chipset	Inference Time (ms)	Peak Memory Range (MB)	Primary Compute Unit
LiteHRNet	TFLITE	float	Snapdragon® 8 Elite Gen 5 Mobile	2.014 ms	0 - 111 MB	NPU

License

- The license for the original implementation of LiteHRNet can be found [here](#).

References

- [Lite-HRNet: A Lightweight High-Resolution Network](#)
- [Source Model Implementation](#)

Community

- Join [our AI Hub Slack community](#) to collaborate, post questions and learn more about on-device AI.
- For questions or feedback please [reach out to us](#).

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